Sagar Mishra

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Summary

Full-stack engineer with about 4 years building TypeScript, Node.js, React and PHP products. Spent two years at Programiz, where the payments, course APIs and certificate engine served 100,000+ paid learners. Has led teams of up to six, and owns work from spec and schema through tests and CI/CD to production. Uses AI coding agents daily and reviews every change before it merges.

Experience

Software Engineer, Jobss.ai, Melbourne *Feb 2026 – May 2026*

- Led a team of four engineers to ship a live Laravel and PHP job and task platform, turning the founder's requirements into tasks and reviewing all 25+ pull requests before merge.
- Built authentication and role-based access control (RBAC) for four user roles, all checked against one permission model.
- Built the employer workspace in Laravel and Blade with one designer and four developers, where employers post jobs and tasks and move applicants through review, shortlist and rejection.
- Sped up the job and task feeds by replacing offset with cursor pagination, so the backend returns deep pages faster. Wrote the logic that allocates available workers to open jobs by demand.

Co-founder & Software Engineer, Iruka Technologies Pvt. Ltd., Kathmandu, Nepal (remote from Melbourne) *Mar 2024 – Jan 2026*
Startup. Development stopped in Jan 2026, before launch.

- Owned architecture, backend, frontend and code review for Nexus (warehouse management) and Aroma (multi-tenant storefront). Led a team of three with a co-founder and a paid intern, building with AI coding tools, and reviewed 400+ pull requests.
- Built the Nexus stock ledger in PostgreSQL with SELECT FOR UPDATE row locks, so two buyers cannot claim the same unit, and tested it with about 10 concurrent checkouts [estimate, verify before sending].
- Cut round trips by moving Aroma from MongoDB to PostgreSQL, so related entities that took several network calls now come back in one join.
- Modelled product drops as a 4-stage state machine (supplier onboarding, lot negotiation, intake, lot-level pricing) and each order as its own state machine, with cron jobs batching retail orders and Redis handing them to Nexus.
- Got product-page LCP under 2 seconds on the Next.js buyer path (search, checkout, order history) by choosing server rendering or static generation per page, with React Query and Zustand.
- Stopped duplicate orders with idempotency keys, and ran the eSewa and Khalti gateways as a separate payment service.

Software Engineer, ParewaLabs (Programiz), Kathmandu, Nepal *Oct 2021 – Nov 2023*

- Built the course-flow REST APIs in Node.js, TypeScript and MySQL, and the React learner dashboard, used by 100,000+ paid learners.
- Integrated Paddle checkout, webhooks and iframe messaging after agreeing the payment flow with the founder, and fixed its live failures reported by Sentry. Every paid learner still pays through it.
- Built the certificate engine in Node.js and TypeScript and fixed its live errors in production. It has issued 100,000+ verified completion certificates.
- Integrated sensAI, an LLM-powered tutor, into the exercise page. Learners ask it about 300 questions a day.
- Built annotations for the online compiler (23M visits a month), and with the principal engineer fixed chunked Python output by adding a socket heartbeat that joins the pieces into one response.
- Delivered Programiz Reports in two weeks, leading a Scrum team of six from the content team's requirements to release, after agreeing its priority with the founder and scoping sprints with the project manager.
- Wrote Playwright tests for the payment modules and Cypress tests for the frontend, in a GitHub Actions CI/CD pipeline that deployed four production apps to DigitalOcean: the course-flow APIs, certificate engine, online compiler and learner dashboard.

Projects

e-warehouse, age-aware lot ledger for clothing merchants (pre-launch). Node.js, TypeScript, Express, PostgreSQL *Sep 2026 – Present*

- Built the backend foundation and wrote 10 architecture decision records. Every API returns one response envelope with fixed error codes, every log line carries a request ID, and every stock change is a new row, never edited in place.
- Merged 58 pull requests from parallel AI coding agents supervised through firstmate, an open-source agent supervisor. Schema, API design, review and every merge stay human.

Agent skills library, 30 private skills for planning, building and reviewing code *Oct 2026 – Present*

- Tests each skill blind against the same task run without it before trusting it; one scored 5 of 5 against 1 of 5. Splitting a 145-rule code skill by topic cut a task's context from about 7,400 to 2,200 tokens.

Skills

Languages: TypeScript, JavaScript, PHP, Python, SQL, Kotlin, C# (university projects)
Frameworks and libraries: Node.js, Express, React, Next.js, React Query, Zustand, Tailwind CSS, Laravel, Django
Tools and platforms: Git, GitHub Actions (CI/CD), DigitalOcean, Vercel, Playwright, Cypress, Selenium, Postman, Sentry, Jira, Claude Code
Databases: PostgreSQL, MySQL, MongoDB, Redis, Drizzle ORM, Sequelize

Education

Master of Information Technology and Systems (Software Engineering), Victorian Institute of Technology, Melbourne *Feb 2024 – Jul 2026*
BSc (Hons) Software Engineering, Softwarica College of IT and E-commerce, Kathmandu, Nepal *Nov 2018 – Oct 2021*